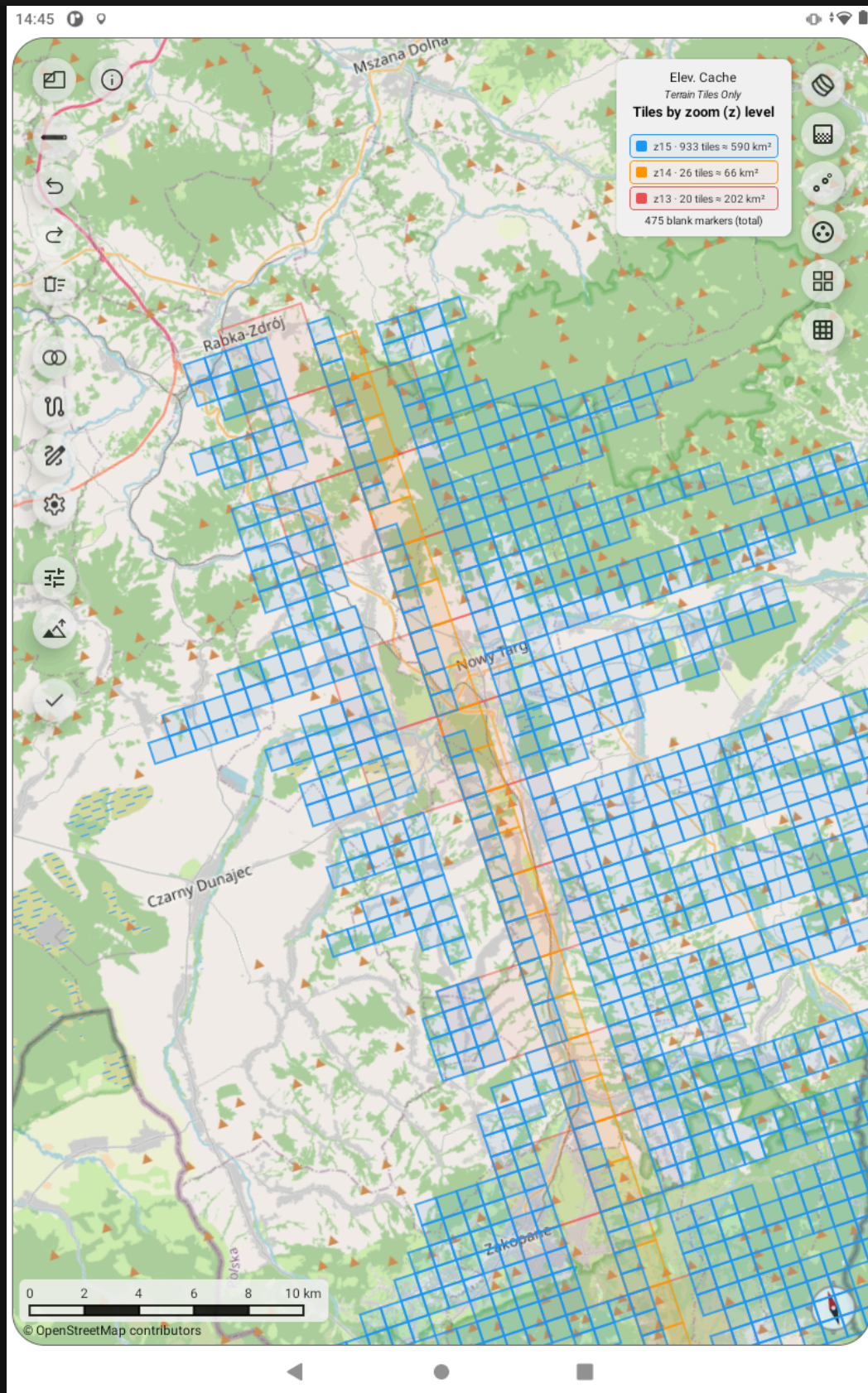


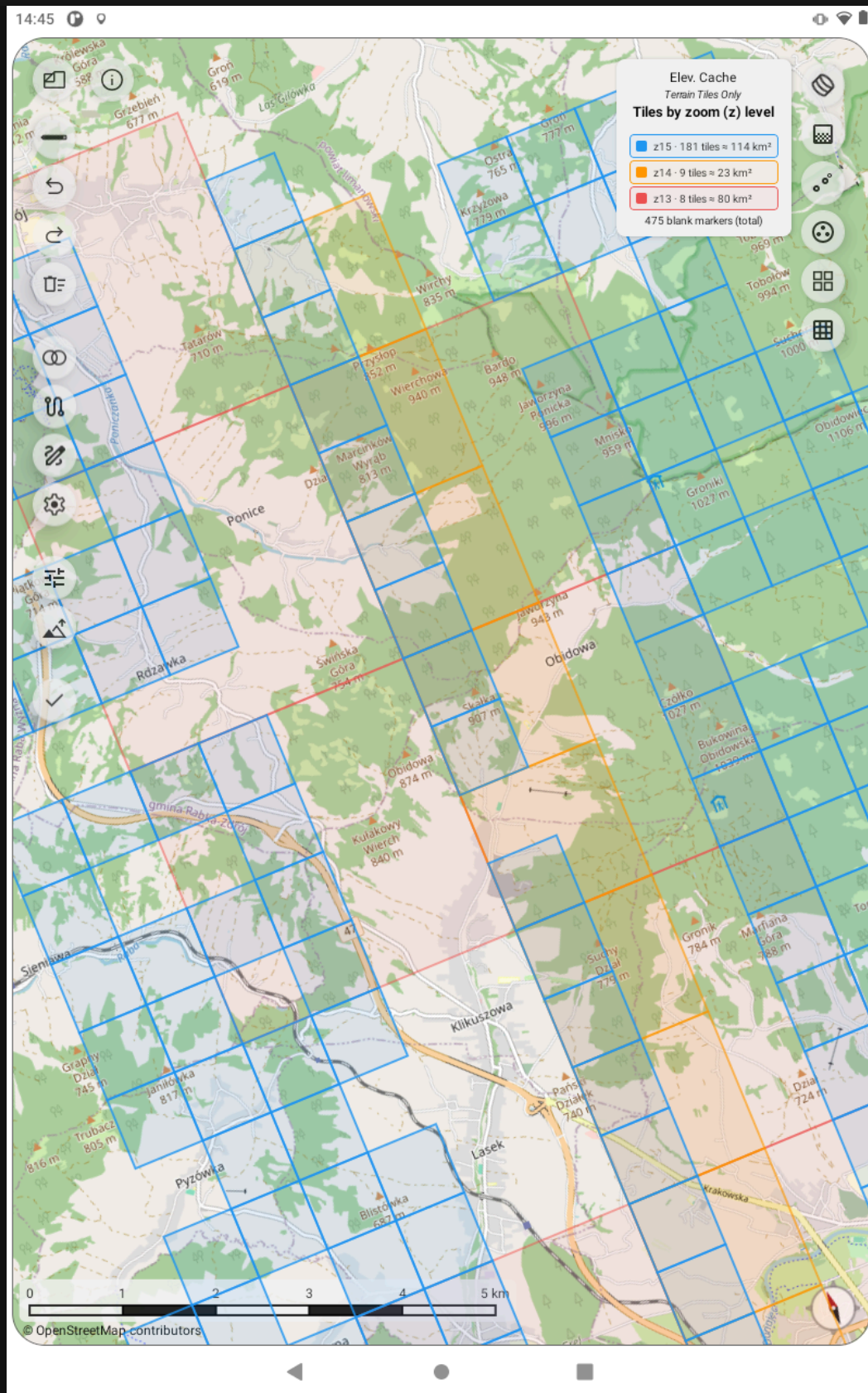


v2.17.0



Multi-Zoom Tile Fallback

AWS terrain tiles have **data voids** at zoom 15.
New cascade: **z15** → **z14** → **z13** fills every gap.
0 API fallbacks needed · 83s wait eliminated.

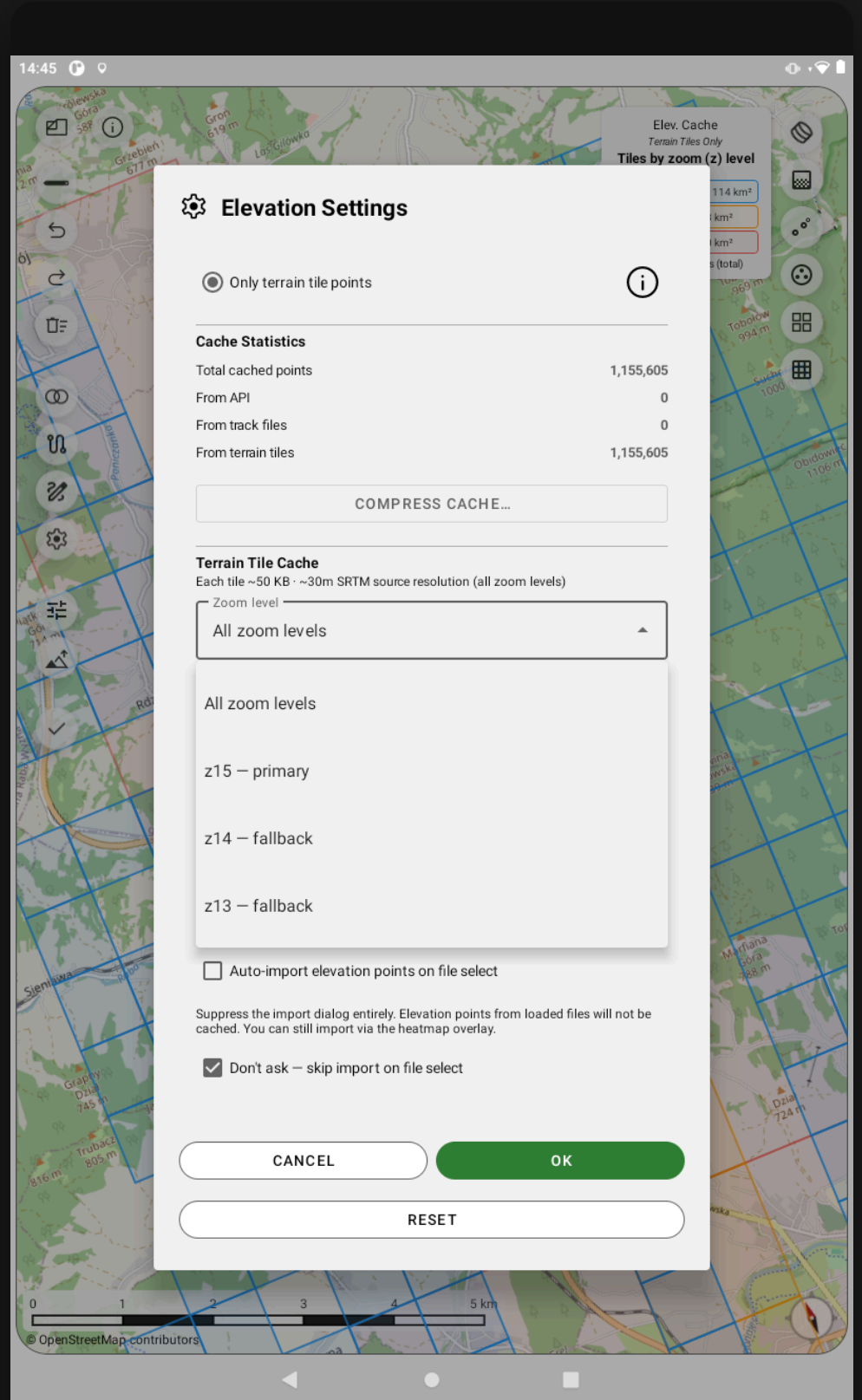
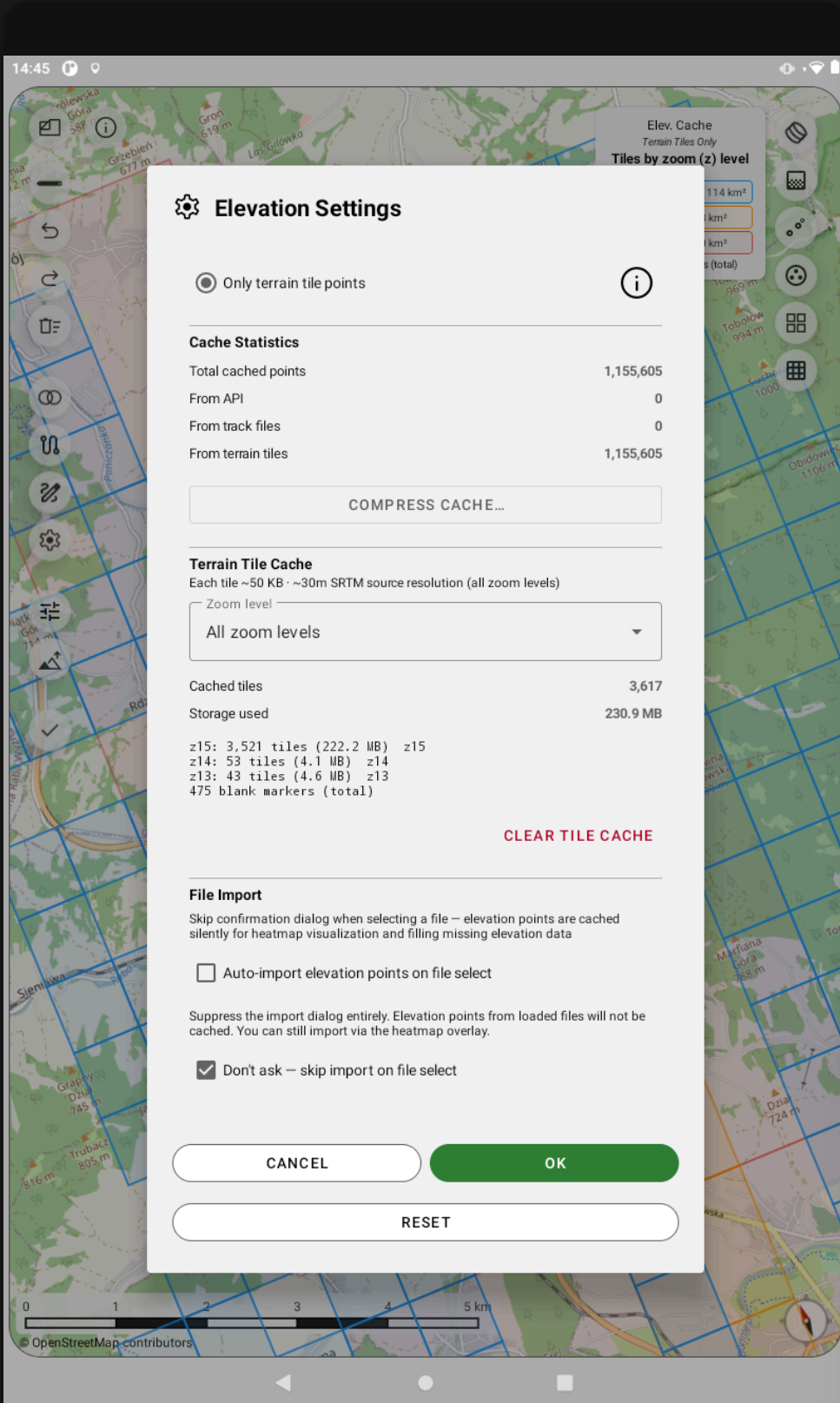


Color-Coded Zoom Boundaries

Tiles by zoom (z) level legend with color-coded cards.

Blue = z15 · **Orange = z14** · **Red = z13** — all ~30m SRTM source.

Per-zoom tile count + area · blank marker tracking.



Per-Zoom Cache Management

Cache breakdown: **z15: 3,521 tiles** · z14: 53 · z13: 61.
Filter by zoom level · clear specific zoom · compress cache.
475 blank markers tracked — no wasted re-fetches.



How the Cascade Works

PHASE 1 — PRIMARY z15 LOOKUP

45,752 points grouped into **1,209 tiles**.

1,129 tiles valid → **39,330 points** resolved.

80 tiles blank → 6,422 NaN points carry forward.

PHASE 2 — z14 PARENT FALLBACK

6,422 NaN points → **~40 parent tiles** fetched on demand.

~300 more points resolved. ~6,122 still NaN.

Blank parents marked with **1-byte sentinel** — never re-fetched.

PHASE 3 — z13 GRANDPARENT FALLBACK

6,122 remaining → **~10 grandparent tiles**.

z13 has **100% coverage** — all points resolved.

Same ~30m SRTM source data — no quality loss.

RESULT — ZERO API MISSES

45,752 / 45,752 points from tiles alone.

API fallback eliminated · **83s wait** → **5s** extra tile fetches.

~2.5 MB extra storage per affected area.



What This Means for You

NO MORE MISSING ELEVATION

Gaps in satellite data used to leave **blank spots** on your heatmap.
Now the app automatically fills them from wider-area tiles.

Same ~30m accuracy — no quality loss.

FASTER ELEVATION LOADING

Previously, missing tiles triggered a **slow online API** fallback.
Now everything resolves from cached tiles locally.

83 seconds of waiting eliminated for typical areas.

WORKS FULLY OFFLINE

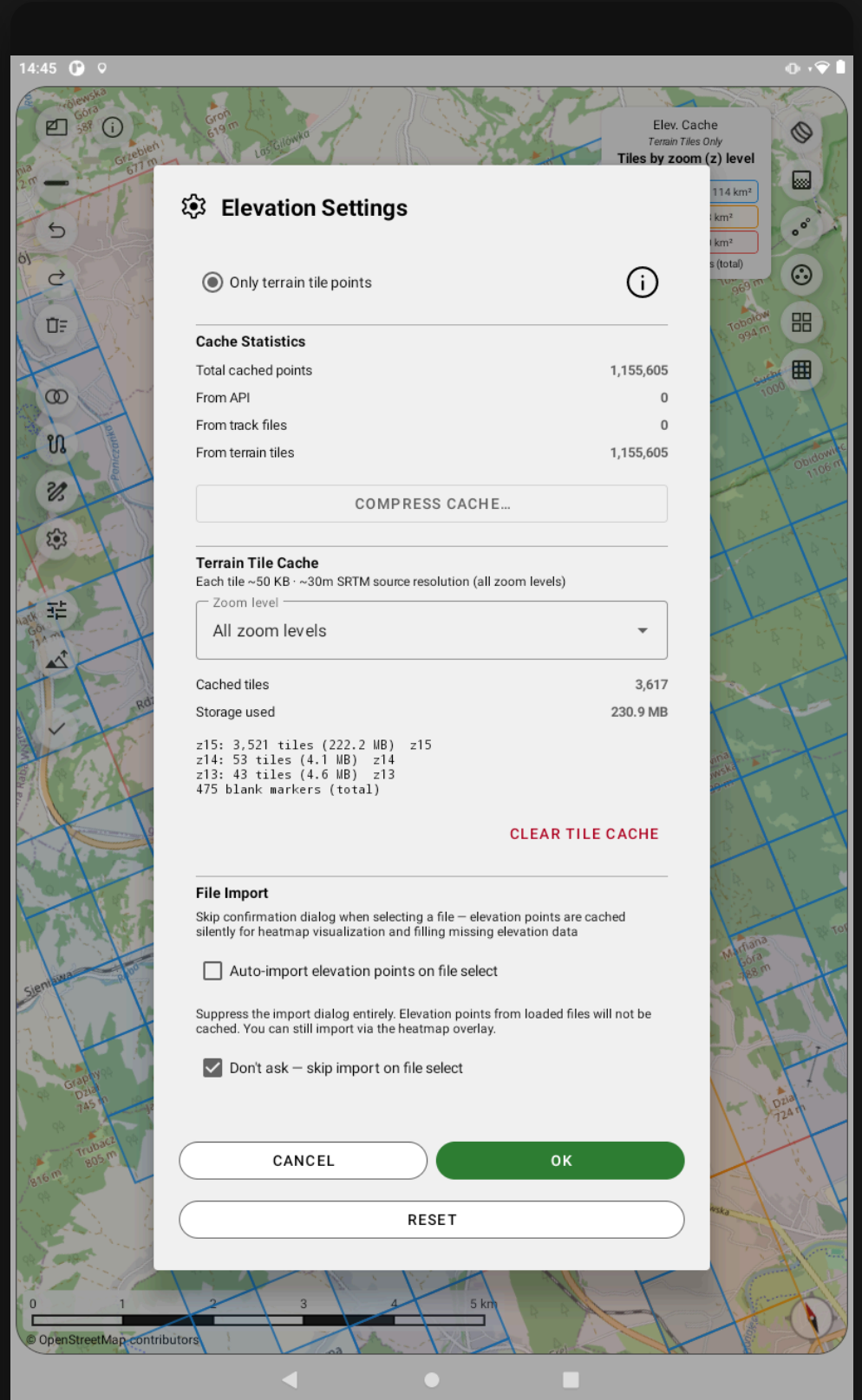
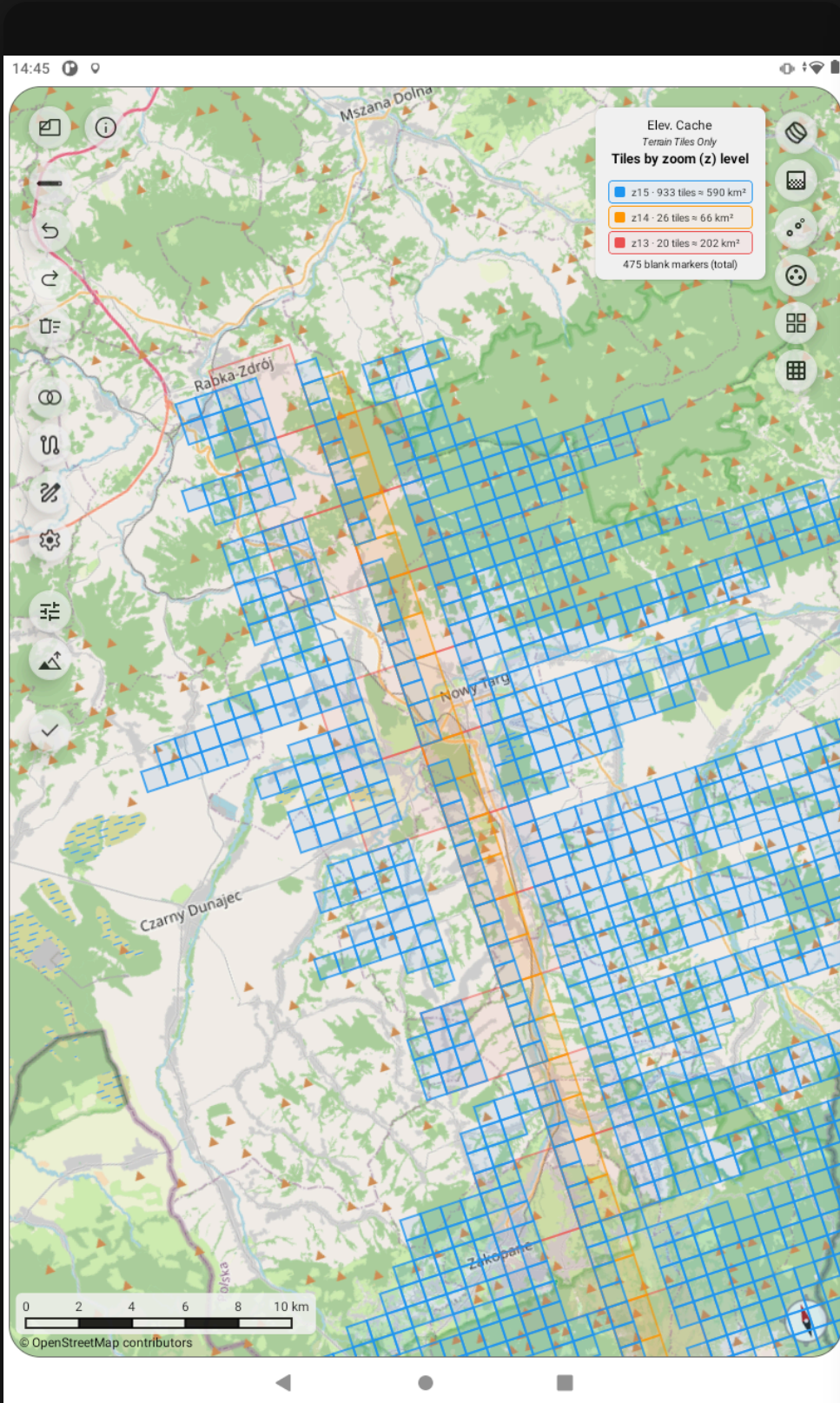
Once tiles are cached, **no internet needed**.

Take your device into the mountains — full elevation data available.
Minimal extra storage: ~2.5 MB per region.

SMART CACHE MANAGEMENT

See exactly how many tiles are cached at each zoom level.
Clear specific zoom levels you no longer need.

Empty tiles tracked — never downloaded twice.



v2.17.0 — Multi-Zoom Tile Fallback

3-level cascade · color-coded boundaries · per-zoom cache
Blank detection · staging writes · void cleanup · viewport rendering

Open to B2B contract or long-term cooperation
Connect or DM → [linkedin.com/in/stanczuk](https://www.linkedin.com/in/stanczuk)